

Microsoft Teams Protection (2020–2021)

Problem: Microsoft Teams data is highly distributed across chats, files, metadata, and permissions, making reliable backup and restore complex at enterprise scale.

What I Did:

- Designed and implemented end-to-end backup, indexing, and restore pipelines for Teams workloads.
- Built support for granular recovery while preserving metadata and relationships.
- Handled complex entity relationships to ensure restore correctness.

Impact: Enabled enterprise-grade protection and recovery for Microsoft Teams, unlocking adoption for large Microsoft 365 tenants.

Public Folders Protection (2020)

Problem: Legacy enterprise customers relied on Public Folders but lacked modern, reliable backup and recovery options.

What I Did:

- Implemented backup and indexing support for deeply nested Public Folder hierarchies.
- Designed restore workflows that correctly rehydrated folder structure and permissions.
- Addressed performance challenges for large legacy datasets.

Impact: Enabled protection and recovery for legacy enterprise mail workloads.

Teams & Groups SaaS Enablement (2022)

Problem: Microsoft 365 workloads needed to be productionized and offered through a scalable, multi-tenant SaaS platform.

What I Did:

- Integrated Teams and Groups protection into a cloud-native SaaS architecture.
- Designed multi-tenant, fault-tolerant workflows with tenant isolation.
- Worked across teams to ensure operational readiness and scalability.

Impact: Made Teams and Groups protection available at scale through Cohesity's SaaS platform.

SharePoint Lists Protection (2023)

Problem: SharePoint Lists contain structured, business-critical data but lacked reliable backup and restore capabilities.

What I Did:

- Designed and shipped backup and indexing support for SharePoint Lists workloads.

- Implemented restore workflows preserving list structure, metadata, and relationships.
- Ensured correctness for large lists with frequent updates.

Impact: Improved data safety and recovery options for structured SharePoint content.

Automated Data Consistency & Self-Healing (2023–2024)

Problem: At large scale, backup systems can drift into inconsistent states, leading to silent restore failures.

What I Did:

- Designed automated mechanisms to detect data inconsistencies across protected datasets.
- Implemented self-healing workflows to safely repair inconsistent states without manual intervention.
- Ensured detection and repair logic was safe, idempotent, and scalable.

Impact: Significantly improved restore reliability at scale; this work resulted in a patent filing.

Permissions Backup & Restore (2024–2025)

Problem: Complex file and folder permissions slowed down backup and restore operations and affected correctness.

What I Did:

- Reworked permission capture logic to reduce redundant processing during backups.
- Redesigned restore workflows to efficiently reapply hierarchical permissions.
- Validated correctness across large datasets with complex permission models.

Impact: Improved backup and restore performance by ~3× while maintaining correctness.

Incremental Refresh (2025)

Problem: Full tenant scans for entity discovery were slow and inefficient for large Microsoft 365 environments.

What I Did:

- Designed incremental refresh mechanisms to track changes instead of full re-scans.
- Implemented efficient state tracking to minimize unnecessary processing.
- Validated behavior across tenants with very large object counts.

Impact: Improved entity discovery speed by ~5× overall and up to ~20× for large tenants.

Scale & Reliability Initiatives (Ongoing)

Problem: Enterprise tenants with 100k+ users exposed scale limits and reliability bottlenecks.

What I Did:

- Drove scale testing to identify performance and reliability issues.
- Led fixes for production issues affecting large enterprise customers.
- Partnered with infrastructure teams to improve system stability under load.

Impact: Enabled consistent backup and restore behavior for very large enterprise tenants.